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Paper 828 — A Memoir on Seminvariants (Arts. 7–36).

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A MEMOIR ON SEMINVARIANTS.

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7. In forming a product $lmn\dots pqr\dots$, we may have equalities among the numbers $l, m, n, \dots$ of the first symbol, and also between the numbers $p, q, r, \dots$ of the second symbol: moreover (whether there are or are not any such equalities), we may have equalities presenting themselves between the numbers $l + p, m + q, \dots$ of any symbol $(l + p)(m + q)\dots$ on the right-hand side of the equation: and the process must be performed so as to take account of all these equalities. The actual process is best shown by an example: say we require the product $3332 \cdot 322$, which is of the deg. weight $6 \cdot 18$.

3332	322	$\div 12$				
322		6	6552	2	1	
32 2		12	6543	1	1	
22 3		6	5553	6	3	
32	2	12	65322	2	2	
3 2	2	6	64332	2	1	
2 3	2	6	55332	4	2	}
22	3	6	55332	4	2	
2 2	3	3	54333	6	3	
3	22	2	632322	12	2	
3	22	1	533322	12	1	}
2	32	6	533322	12	6	
2	32	2	433332	36	6	
	322	1	3333222	144	12	
		73				

8. Observe first that $\{4, 3\} = 24 + 36 + 12 + 1, = 73$. In placing all or any of the numbers of the 322 under those of the 3332, we do this in all the really distinct ways, inserting a numerical coefficient for the frequency of each way. Thus when the whole of the 322 is thus placed, the 3 may be under a 3, and the two 2's may be under 33 or under 32; or else the 3 may be under a 2, and the two 2's must then be under 33: there are thus three ways, and these have the frequencies 6, 12, 6 respectively. For as to the first way, the 3 may be under any one of the three 3's, and for each such position of the 3, the 22 can be placed in either of two orders under the other two 3's; the frequency is thus $3 \cdot 2, = 6$. And similarly the other two frequencies are 12 and 6; the sum $6 + 12 + 6, = 24$, is, the first term of $\{4, 3\}$. In like manner, when two of the numbers of the 322 are placed under the 3332, there are five ways having the frequencies 12, 6, 6, 6, 6 respectively, the sum of these frequencies is 36, which is the second term of $\{4, 3\}$. And so when one number of the 322 is placed under the 3332, there are four ways having the frequencies 3, 1, 6, 2 respectively: the sum of these frequencies is 12, which is the

third term of $\{4, 3\}$. Finally, when no number of the 322 is placed under the 3332, there is a single way, having the frequency 1, which is the last term of $\{4, 3\}$. These agreements are a useful verification for the frequencies: the sum of all the frequencies is of course = 73, the value of $\{4, 3\}$.

9. We next form the column of symmetric functions by adding to each line the 3332: to avoid accidental errors of addition, it is proper to verify for each of these that the weight is the sum, = 18, of the weights 11 and 7 of the two factors respectively. It is to be observed that the same symmetric function may present itself more than once: thus we have the functions 55332, and 533322, each of them twice. We then form a column of multiplicities: in 5553, the three 5's give 6: in 63322, the two 3's and the three 2's give $2 \cdot 6$, = 12, and so on. There is in like manner for the factor 3332 a multiplicity 6, and for the factor 322 a multiplicity 2; and these combined together give $6 \cdot 2$, = 12, viz. this is the heading $\div 12$ of the column. And then forming the products $6 \cdot 2$, $12 \cdot 1$, $6 \cdot 6$, &c., of the corresponding frequencies and multiplicities and dividing in each case by 12, we have the right-hand column of numerical coefficients.

10. The result is to be read

$$3332 \cdot 322 = 1 \cdot 6552 + 1 \cdot 6543 + 3 \cdot 5553 + \&c.,$$

the coefficients 2, 2 and 1, 6 of the repeated terms being of course united together, so that we have

$$+ 4 \cdot 55332 \quad + 7 \cdot 533322.$$

With a little practice, the operation is performed without difficulty and with very small risk of error.

11. We may apply the process to obtain analytical formulae for certain forms of products. Consider, for instance, the product $2^\alpha \cdot 2^\beta$, where $\alpha > \beta$ or $= \beta$: and in the coefficients write for shortness α , instead of $[\alpha]^\alpha$ or $\Pi\alpha$, to denote $1 \cdot 2 \cdot 3 \dots \alpha$. We have

$$\frac{2^\alpha | 2^\beta}{2^A | 2^{\beta-A}} = \frac{\alpha}{A \cdot \alpha - A \cdot \beta - A} \cdot 4^A 2^{\alpha+\beta-2A}, \quad A \cdot \alpha + \beta - 2A \cdot \frac{\alpha + \beta - 2A}{\alpha - A \cdot \beta - A},$$

viz. the formula is

$$2^\alpha \cdot 2^\beta = \Sigma \frac{\alpha + \beta - 2A}{\alpha - A \cdot \beta - A} \cdot 4^A 2^{\alpha+\beta-2A},$$

where A has any integer value from β to 0; the first term is $= 1 \cdot 4^\beta 2^{\alpha-\beta}$, and the last term is $\frac{\alpha + \beta}{\alpha \cdot \beta} \cdot 2^{\alpha+\beta}$. The weight of any term is $4A + 2(\alpha + \beta - 2A)$, = $2\alpha + 2\beta$, as it should be.

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In explanation as to the frequency, observe that out of the β 2's we take any A 2's and place them in any order under any A of the α 2's. The number of combinations taken is $\frac{\beta}{A \cdot \beta - A}$, which (in A orders) gives $\frac{\beta}{\beta - A}$ sets to be placed under $\frac{\alpha}{A \cdot \alpha - A}$ sets out of the α 2's: we thus have the foregoing coefficient of frequency $\frac{\alpha}{A \cdot \alpha - A} \cdot \frac{\beta}{\beta - A}$, where as mentioned above α , &c. are written to denote $[\alpha]^\alpha$, &c.

12. We may in like manner find a formula for the product $3^\alpha 2^\beta \cdot 3^\gamma 2^\delta$. Taking $\gamma, = x + y + z$, any partition of γ , and $\delta, = p + q + r$, any partition of δ , and writing also $A = x$, $B = y + p$, $C = q$, we have

$$\frac{3^\alpha 2^\beta \mid 3^\gamma 2^\delta}{3^A 2^B 3^C 2^E \mid 3^B 2^C \mid 6^A 5^B 4^C 3^{\alpha+\gamma-2A-B} 2^{\beta+\delta-B-2C} \mid N}$$

where for the frequency, M , we have

$$\begin{aligned} \text{No. of terms } 3^x &= \frac{\gamma}{x \cdot \gamma - x}, & 3^y &= \frac{\gamma - x}{y \cdot z}, \\ , & & 2^p &= \frac{\delta}{p \cdot \delta - p}, & 2^q &= \frac{\delta - p}{q \cdot r}, \end{aligned}$$

to be placed under

$$\begin{aligned} \text{sets of terms } 3^x &= \frac{\alpha}{x + p \cdot \alpha - x - p}, & 2^p &= \frac{\beta}{y + q \cdot \beta - y - q}, \\ \text{in orders, } &= x + p; & &= y + q, \end{aligned}$$

whence, multiplying, we have for the frequency

$$M = \frac{\alpha \cdot \beta \cdot \gamma \cdot \delta}{x \cdot y \cdot z \cdot p \cdot q \cdot r \cdot \alpha - x - p \cdot \beta - y - q},$$

and for the multiplicity, N , we have

$$\begin{aligned} N &= A \cdot B \cdot C \cdot \alpha + \gamma - 2A - B \cdot \beta + \delta - B - 2C, \\ &= x \cdot B \cdot y \cdot \alpha + \gamma - 2A - B \cdot \beta + \delta - B - 2C. \end{aligned}$$

13. The coefficient is thus

$$\begin{aligned} \frac{M \cdot N}{\alpha \cdot \beta \cdot \gamma \cdot \delta} &= \frac{B \cdot x + y - 2A - B \cdot \beta + \delta - B - 2C}{y \cdot z \cdot p \cdot r \cdot \alpha - x - p \cdot \beta - y - q}, \\ &= \frac{B \cdot x + y - 2A - B \cdot \beta + \delta - B - 2C}{y \cdot p \cdot \gamma - A - y \cdot \alpha - A - p \cdot \beta - C - y \cdot \delta - C - p}, \end{aligned}$$

or, putting also

$$\begin{aligned} D &= \alpha + \gamma - 2A - B, \\ E &= \beta + \delta - B - 2C, \end{aligned}$$

whence

$$6A + 5B + 4C + 3D + 2E = 3(\alpha + \gamma) + 2(\beta + \delta),$$

the formula finally is

$$3^\alpha 2^\beta \cdot 3^\gamma 2^\delta = \Sigma \Lambda \cdot 6^A 5^B 4^C 3^D 2^E,$$

where

$$\Lambda = \frac{B}{y \cdot p \cdot \gamma - A - y} \cdot \frac{D}{\alpha - A - p \cdot \beta - C - y} \cdot \frac{E}{\delta - C - p},$$

or, as this may also be written,

$$= \frac{B}{y \cdot D - y} \cdot \frac{D}{z \cdot D - z} \cdot \frac{E}{r \cdot E - r},$$

so that the coefficient Λ is in fact the product of three binomial coefficients: it must, however, be recollected that the same term $6^A 5^B 4^C 3^D 2^E$ may occur more than once, with different coefficients Λ , so that in the final result, when these terms are united together, the numerical coefficients are not each of them of the form in question.

14. The limits of the summation are conveniently defined by means of the diagram

A	p	$\alpha - A - p$	α	$D = \alpha + \gamma - 2A - B,$
y	C	$\beta - C - y$	β	$E = \beta + \delta - B - 2C,$
z	r			<i>ut suprâ;</i>
γ	δ			

viz. here the sums of the first and second lines are α and β respectively, and the sums of the first and second columns are γ and δ respectively; we have $B = y + p$, a partition of B ; but the values of y, p must be such as not to render negative any one of the four terms $z, r, \alpha - A - p, \beta - C - y$ of the diagram: for if any one of these numbers were negative, the corresponding factorial in the denominator of Λ would be infinite and we should have $\Lambda = 0$. For any given term $6^A 5^B 4^C 3^D 2^E$ the proper weight $3(\alpha + \gamma) + 2(\beta + \delta)$, there may be no suitable values of y, p , and the coefficient is then $= 0$: there may be a single pair of values, and the coefficient is then (as remarked above) a product of three binomial coefficients: or there may be more than a single pair of values, and the entire coefficient of the term has not in this case a like simple form.

15. To exhibit the working of the formula, I apply it to the recalculation of the foregoing product $3332 \cdot 322$ ($\alpha, \beta, \gamma, \delta = 3, 1, 1, 2$). Properly the whole series of symbols 666, 6642, 6633, &c., of the symmetric functions of the weight 18 should be written down, and the coefficient be calculated for each of them: but I write down only those which have coefficients not $= 0$: for each of these, I take *all* the partitions

$B = y + p$, several of these giving, as will be seen, zero values, and the others giving the values already obtained for the coefficients of the several terms.

[The original page presents a wide working-table, in which each row lists a symmetric-function symbol on the left, then the five exponents A, B, C, D, E , then a partition (B, y, p) , then three groups of binomial-coefficient data (numerators divided by denominators for the three factors of Λ in Art. 13), and a final column showing the actual contribution. The summary table below gives the exponent-set and the final contribution for each row; the intermediate binomial-coefficient arithmetic, being purely numerical, is suppressed.]

Symbol	A	B	C	D	E	Contribution
6552	1	2	0	0	1	$1 \cdot 6552$
6543	1	1	1	1	0	$1 \cdot 6543$
6532 ²	1	1	0	1	2	$1 \cdot 6532^2$
64332	1	0	1	2	1	$1 \cdot 64332$
6332 ²	1	0	0	2	3	$1 \cdot 6332^2$
5553	0	3	0	1	0	$3 \cdot 5553$
5532 ²	0	2	0	2	1	$\} 4 \cdot 55332$
54333	0	1	1	3	0	$1 \cdot 54333$
5332 ³	0	1	0	3	2	$\} 7 \cdot 5333^2$
433332	0	0	1	4	1	$4 \cdot 433332$
33332 ²	0	0	0	4	3	$11 \cdot 33332^2$

The entries grouped by $\}$ (the two rows for 5532²/5532² and the two for 5332³/5332³) correspond to two partitions giving the same final symbol whose coefficients combine.

It is quite possible that abbreviations and verifications might be introduced, but the process as it stands seems to be at once less expeditious and less safe than the one first made use of.

16. Particular cases of the general formula are

$$2^\alpha \cdot 2^\beta = \Sigma \Lambda 4^C 2^E; \quad E = \beta + \delta - 2C, \quad \text{and thence } 4C + 2E = 2(\beta + \delta),$$

$$\Lambda = \frac{E}{\beta - C \cdot \delta - C};$$

which agrees with a result before obtained. Again,

$$3^\alpha 2^\beta \cdot 2^\delta = \Sigma \Lambda \cdot 5^B 4^C 3^D 2^E; \quad D = \alpha - B,$$

$$E = \beta + \delta - B - 2C,$$

and thence

$$5B + 4C + 3D + 2E = 3\alpha + 5B + 4C + 3D + 2E = 3\alpha + 2\beta + 2\delta,$$

$$\Lambda = \frac{E}{\beta - C \cdot \delta - B - C}.$$

17. We may, in like manner with the formula for $3^\alpha 2^\beta \cdot 3^\gamma 2^\delta$ obtain the following:

$$4^\beta 3^{2\alpha} \cdot 2^\delta = \Sigma \Lambda \cdot 6^A 5^B 4^C 3^D 2^E,$$

where

$$D = \alpha - B, \quad E = 2\beta + \beta + \delta - 3A - B - 2C,$$

and thence

$$6A + 5B + 4C + 3D + 2E = 4\beta + 3\alpha + 2\beta + 2\delta,$$

and the value of the coefficient Λ is

$$\Lambda = \frac{C}{\beta - A \cdot C - \beta} \cdot \frac{E}{\alpha + \beta - A - C \cdot \beta + \delta - 2A - B - C},$$

viz. Λ is the product of two binomial coefficients: and since here a given term $6^A 5^B 4^C 3^D 2^E$ occurs once only, each numerical coefficient is actually of this form.

In the particular case $\delta = 0$, we must have $A = 0$ (this appears *a posteriori* from the denominator factor $-A$, a factorial which is infinite for any positive value of A); and we thus obtain the first given formula for $3^\alpha 2^\beta \cdot 2^\delta$.

Capitation and Decapitation. Art. Nos. 18 to 21.

18. I explain the converse processes of Decapitation and Capitation. In any symmetric function, for instance 6552 of the degree 6, the whole coefficient of α^6 is 552, this symbol referring in the first instance to the series of remaining roots $\beta, \gamma, \delta, \dots$; but as the series of roots is unlimited, we may ultimately replace these by $\alpha, \beta, \gamma, \dots$, and so use 552 in its original sense. Similarly, in 6652, the whole coefficient

of α^6 is 652, the only difference being that, while in the former case the degree is reduced to 5, in the present case it remains = 6. In every case we decapitate the symbol by striking out the highest number—in the case of two or more equal numbers, one only of these being struck out. Observe that by decapitation we always diminish the weight, but we do or do not diminish the degree. In a product such as $3332 \cdot 322$ we obtain in like manner the whole coefficient of α^6 by the decapitation of each factor, viz. the coefficient is = $332 \cdot 22$: and in any equation such as that obtained above for the product $3332 \cdot 322$, the whole coefficient of the highest power of α must be = 0, viz. we can by decapitation obtain a new equation of lower weight: thus from the equation in question of weight 18, we obtain the new equation of weight 12,

$$\begin{aligned} 332 \cdot 22 &= 1 \cdot 552 \\ &+ 1 \cdot 543 \\ &+ 2 \cdot 5322 \\ &+ 1 \cdot 4332 \\ &+ 3 \cdot 33222, \end{aligned}$$

where observe that the terms of a degree lower than 6 in the original equation give no term in the new equation. The new equation of the deg. weight $5 \cdot 12$ might of course be obtained independently in like manner with the original equation.

19. We capitate a symbol by prefixing to it a number which is not less than the highest number contained in it: thus 552 may be capitated into 5552, 6552, &c.; and so a product $332 \cdot 22$ may be capitated into $3332 \cdot 222$, $3332 \cdot 322$, &c.; moreover a single symbol may be capitated into a product, 552 into $5552 \cdot 4$: in fact, the capitation may be any operation such that by decapitation we reproduce the original symbol. The increase of weight may be any number not less than the degree of the original symbol: but it is usually taken to be a given number: thus for any symbol of a degree not exceeding 6, we may capitate so as to increase the weight by 6. The capitation does or does not increase the degree.

20. An identical equation may be capitated in a variety of ways, but instead of an identity we obtain only a congruence, that is, an equation which requires to be completed by the adjunction to it of proper terms of lower degrees. Thus from the above equation of the deg. weight $5 \cdot 12$ we may obtain

$$\begin{aligned} 3332 \cdot 322 &\equiv 1 \cdot 6552 \\ &+ 1 \cdot 6543 \\ &+ 2 \cdot 65322 \\ &+ 1 \cdot 64332 \\ &+ 3 \cdot 633222. \end{aligned}$$

Imagine here all the terms brought to the same side so that the form is $\Omega \equiv 0$: Ω is a function not containing α^6 , for the whole coefficient of α^6 therein is precisely

that function which by the equation of the deg. weight $5 \cdot 12$ is expressed to be $= 0$: and hence Ω , *qua* symmetric function cannot contain $\beta, \gamma, \delta, \dots$; viz. Ω is a symmetric function of the degree 5 at most: the congruence $\Omega \equiv 0$ thus means, $\Omega =$ a properly determined function of the degree 5 at most. Obviously by development of the term $3332 \cdot 322$, that is, by substituting for this term its value as given by the equation of the deg. weight $6 \cdot 18$, the function of the degree 5 at most would be found to be $=$ the sum of those terms in the expression of $3332 \cdot 322$, which are of a degree inferior to 6: and the congruence $\Omega \equiv 0$ thus completed would be nothing else than the equation of the deg. weight $6 \cdot 18$.

21. We might have capitated in a different manner: for instance

$$\begin{aligned} 4332 \cdot 222 &\equiv 1 \cdot 6552 \\ &+ 1 \cdot 6543 \\ &+ 2 \cdot 65322 \\ &+ 1 \cdot 44332 \cdot 2 \\ &+ 3 \cdot 33222 \cdot 3, \end{aligned}$$

there being here three products requiring to be developed: on replacing them by their values, there would be found for Ω a value which would be a determinate function of a degree less than 6: and putting $\Omega =$ this value, the congruence $\Omega \equiv 0$ would be completed into an equation.

Seminvariants of a given Degree; Perpetuants, &c. Art. Nos. 22 to 38.

22. We consider now seminvariants according to their degrees; in particular, those of the degrees 2, 3, 4, 5 and 6, or say quadric, cubic, quartic, quintic and sextic seminvariants: the forms of these are $2^{\alpha+1}, 3^{\alpha+1}2^{\beta}, 4^{\alpha+1}3^{\beta}2^{\gamma}, 5^{\alpha+1}4^{\beta}3^{\gamma}2^{\delta}, 6^{\alpha+1}5^{\beta}4^{\gamma}3^{\delta}2^{\epsilon}$, where each exponent $\alpha, \beta, \gamma, \delta, \epsilon$ is a positive integer not excluding zero: the exponent of the highest number is in each case written $\alpha + 1$, and has thus the value 1 at least, for otherwise the form would not be of the proper degree. The several weights are $2(\alpha + 1), 3(\alpha + 1) + 2\beta, 4(\alpha + 1) + 3\beta + 2\gamma$, &c., or, what is the same thing, we have for the several degrees respectively

$$\begin{aligned} w - 2 &= 2\alpha, \\ w - 3 &= 3\alpha + 2\beta, \\ w - 4 &= 4\alpha + 3\beta + 2\gamma, \\ w - 5 &= 5\alpha + 4\beta + 3\gamma + 2\delta, \\ w - 6 &= 6\alpha + 5\beta + 4\gamma + 3\delta + 2\epsilon, \end{aligned}$$

and we have for a given degree and weight as many seminvariants as there are systems of exponents satisfying the corresponding equation.

23. These numbers are at once expressible by means of a series of Generating Functions (G.F.), viz. writing for shortness 2, 3, 4, 5, 6 to denote $1 - x^2$, $1 - x^3$, $1 - x^4$, $1 - x^5$, $1 - x^6$, the G.F.'s are

$$\frac{x^2}{2}; \quad \frac{x^3}{2 \cdot 3}; \quad \frac{x^4}{2 \cdot 3 \cdot 4}; \quad \frac{x^5}{2 \cdot 3 \cdot 4 \cdot 5}; \quad \frac{x^6}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6}.$$

In fact, the number of seminvariants of a given weight is = coefficient of x^w in the corresponding G.F.; for the quadric seminvariants (or those of deg. weight $2 \cdot w$) in $\frac{x^2}{2}$; for the cubic seminvariants (or those of deg. weight $3 \cdot w$) in $\frac{x^3}{2 \cdot 3}$; and so for the others.

24. A seminvariant of a given degree may be a sum of products (of that degree) of seminvariants of lower degrees, and of seminvariants of lower degrees: and it is in this case said to be reducible: a seminvariant which is not reducible is said to be irreducible, or otherwise to be a *perpetuant*. This notion of a perpetuant is due to Sylvester, see his Memoir “On Subinvariants, *i.e.* Semi-Invariants to Binary Quantics of an Unlimited Order,” *American Journal of Mathematics*, vol. v. (1882–83), pp. 78–137 (§ 4 Perpetuants, pp. 105–118). In speaking of the number of perpetuants of a given deg. weight, we assume throughout that these are independent perpetuants, not connected by any linear relation.

25. Since the seminvariants used for the reduction of a given reducible seminvariant can themselves be expressed in terms of perpetuants, we may say more definitely that a seminvariant of a given degree, which is a sum of products (of that degree) of perpetuants of lower degrees, and of perpetuants of lower degrees, is reducible. The words “of that degree” are essential to the definition: a seminvariant may be expressible as a sum of products (of a higher degree) of perpetuants of lower degrees, and of perpetuants of lower degrees, and it is not on this account reducible: a seminvariant so expressible is said to be a “syzygant”: but as to this, see No. 49.

26. Every quadric or cubic seminvariant is obviously a perpetuant: the quadric and cubic perpetuants have thus the before-mentioned G.F.'s $\frac{x^2}{2}$ and $\frac{x^3}{2 \cdot 3}$ respectively.

27. A reducible quartic seminvariant can only be a sum of products ($2 \cdot 2$) of two quadric perpetuants, and of quadric perpetuants, and it is clear that no quartic seminvariant the symbol of which contains a 3 is thus expressible. If the symbol does not contain a 3, viz. when the form is $4^{\alpha+1}2^{\gamma}$, the seminvariant is reducible: we have for instance

$$4 = 2 \cdot 2 - 2 \cdot 22, \\ 42 = 22 \cdot 2 - 3 \cdot 222, \quad \&c.$$

To show that this is so in general, observe that any symmetric function $2^{\alpha+1}1^{\gamma}$, *qua* symmetric function can be expressed as a rational and integral function of the

degree $2(\alpha + 1) + \gamma$, of the coefficients $1, 1^2, 1^3$, &c.: instead of the roots considering their squares, we have thence an expression for the quartic seminvariant $4^{\alpha+1}2^\gamma$ in terms of the quadric perpetuants $2, 2^2, 2^3$, &c., and such expression will be of the same degree $4(\alpha + 1) + 2\gamma$, as the quartic seminvariant.

It thus appears, as regards the quartic seminvariants, that whenever the symbol contains a 3, and in this case only, the seminvariant is a perpetuant: or, what is the same thing, the form of a quartic perpetuant is $4^{\alpha+1}3^{\beta+1}2^\gamma$: for the weight w , the number is equal to that of the sets of values α, β, γ , such that $w - 7 = 4\alpha + 3\beta + 2\gamma$; or, what is the same thing, the G.F. of the quartic perpetuants is $= \frac{x^7}{2 \cdot 3 \cdot 4}$.

28. Sylvester, in the memoir referred to, obtained this result in a different manner: the quartic seminvariants of a given weight are the quartic perpetuants of that weight and also the products (of that weight) of two quadric perpetuants, the same or different: say (4) is the G.F. for the perpetuants, and (2, 2) for the products: then the G.F. for the quartic seminvariants being as already mentioned $= \frac{x^4}{2 \cdot 3 \cdot 4}$, we have his equation

$$(4) + (2, 2) = \frac{x^4}{2 \cdot 3 \cdot 4}.$$

He deduces (2, 2) from the G.F. $= \frac{x^2}{2}$ of the quadric perpetuants and thence obtains (4), $= \frac{x^7}{2 \cdot 3 \cdot 4}$, as above.

29. Write for a moment ϕx to represent the G.F. $\frac{x^2}{2}$ of the quadric perpetuants, and $A, B, C, \dots$ to represent these quadric perpetuants: we have, in an algorithm which will be readily understood,

$$\begin{aligned}\phi x &= (A + B + C + \dots), \\ (\phi x)^2 &= (A + B + C + \dots)^2 = A^2 + 2AB + \dots, \\ \phi x^2 &= A^2 + \&c.;\end{aligned}$$

and thence

$$\frac{1}{2}\{(\phi x)^2 + \phi x^2\} = A^2 + AB + \&c.;$$

viz. the G.F. (2, 2), is

$$\frac{1}{2}\{(\phi x)^2 + \phi x^2\} = \frac{1}{2}\left\{\frac{x^4}{2 \cdot 2} + \frac{x^4}{2 \cdot 4}\right\} = \frac{1}{2}\left\{\frac{x^4(1 + x^2)}{2 \cdot 4} + \frac{x^4(1 - x^2)}{2 \cdot 4}\right\} = \frac{x^4}{2 \cdot 4},$$

and we thence have

$$(4) = \frac{x^4}{2 \cdot 3 \cdot 4} - \frac{x^4}{2 \cdot 4} = \frac{x^7}{2 \cdot 3 \cdot 4},$$

that is, (4) $= \frac{x^7}{2 \cdot 3 \cdot 4}$, the same result as was found above by independent considerations.

30. Sylvester established in like manner (but without the terms S which will be presently explained) the equations

$$(3) + (3, 2) = \frac{x^5}{2 \cdot 3 \cdot 4 \cdot 5} + S_5,$$

$$(6) + (4, 2) + (3, 3) + (2, 2, 2) = \frac{x^6}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + S_6,$$

viz. here (5) is the G. F. for the quintic perpetuants, (2, 3) that for the products of the quadric and the cubic perpetuants: and similarly (6) is the G. F. for the sextic perpetuants, (4, 2) that for the products of the quadric and the quartic perpetuants, (3, 3) for the products of two cubic perpetuants, the same or different: and (2, 2, 2) for the products of three quadric perpetuants, the same or different. We have at once $(3, 2) = (3) \cdot (2)$; $(4, 2) = (4) \cdot (2)$; (3, 3) is found by the same process as was used for finding (2, 2), substituting only $\frac{x^3}{3}$ for $\frac{x^2}{2}$; and (2, 2, 2) is found by a like process, viz. the G. F. for $A^3 + A^2B + ABC + \dots$

is $\frac{1}{6}\{(\phi x)^3 + 3\phi x \cdot \phi x^2 + 2\phi x^3\}$, where $\phi x = \frac{x^2}{2}$. Reducing to the common denominator $2 \cdot 4 \cdot 6$, the numerator is

$$= \frac{1}{6}x^6\{(1+x^2)(1+x^2+x^4) + 3(1-x^4) + 2(1-x^2)(1-x^4)\}.$$

viz. this is x^6 . The several functions thus are

$$(3, 2) = \frac{x^5}{2 \cdot 2 \cdot 3}, \quad (4, 2) = \frac{x^6}{2 \cdot 3 \cdot 4},$$

$$(3, 3) = \frac{x^6}{3 \cdot 6}, \quad (2, 2, 2) = \frac{x^6}{2 \cdot 4 \cdot 6}.$$

31. Mr Hammond, in regard to the equation for the quintic perpetuants, made the very important observation—see his paper “On the Solution of the Differential Equation of Sources,” *American Journal of Mathematics*, t. v. (1882), pp. 218–227—that the products (3, 2) of a cubic perpetuant and a quadric perpetuant are not independent: we have between them syzygies such as $32 \cdot 2 - 3 \cdot 22 = 0$ (viz. the difference of the two products contains no term of the degree 5; the actual value is $= 43 + 322$: Hammond’s equation (12), p. 222), hence the necessity in the equation of a term S_5 referring to these syzygies; and he moreover obtained the expression, $S_5 = \frac{x^7}{2 \cdot 4}$, of the G. F. for these syzygies.

The equation gives

$$(5) = \frac{-x^5 + x^6 + x^{10} + x^{15}}{2 \cdot 3 \cdot 4 \cdot 5} + S_5,$$

where of course the first term is the value of (5) which would be given by the equation without the term S_5 ; and substituting herein for S_5 the foregoing value, viz.

$$S_5 = \frac{x^7}{2 \cdot 4} = \frac{x^7(1-x^2)(1-x^4)}{2 \cdot 3 \cdot 4 \cdot 5},$$

we find

$$(5) = \frac{x^{15}}{2 \cdot 3 \cdot 4 \cdot 5},$$

which is the correct value of the G. F. for the quintic perpetuants: the lowest quintic perpetuant is thus of the weight 15.

32. The equation for the sextic perpetuants gives

$$(6) = \frac{-x^6 + x^{10} + 2x^{16} + x^{15}}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + S_6,$$

which is an equation connecting (6) and S_6 , the G. F.'s for the sextic perpetuants, and the sextic syzygies respectively. I have, in the investigation of the value of S_6 , met with a difficulty which I have not been able to overcome: but I find that

$$(6) = \frac{x^6 + x^{10} + 2x^{15} - x^{16} - x^{18} + \omega(x)}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6},$$

where $\omega(x)$ is possibly the monomial function x^{31} , but this result (which Capt. MacMahon believes to be true) is not yet completely established; it is a function containing no term lower than x^{31} . We have therefore

$$(6) = \frac{\omega(x)}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6},$$

and there is, it would appear, no sextic perpetuant of a weight lower than 31.

33. But before entering on the investigation, it is proper to further develop the theory of the quintic perpetuants. We have quintic seminvariants: 5 for weight 5; 52 for weight 7; 53 for weight 8; 54, 522 for weight 9; 55, 532 for weight 10; 542, 533, 5222 for weight 11; and so on. These are reduced by means of the products $3 \cdot 2$ of a cubic perpetuant and a quadric perpetuant; viz. for any given weight, we have all the products $3^a \cdot 2^b$ of that weight. Thus for weight 5 there is only the product $3 \cdot 2$, and this in fact serves to reduce the seminvariant 5: we have

$$3 \cdot 2 = 5 + 32, \quad \text{and therefore} \quad 5 = 3 \cdot 2 - 32.$$

For the weight 7, there is only the seminvariant 52, and there are the two products $32 \cdot 2$ and $3 \cdot 22$: either of these would serve for the reduction: we have

$$32 \cdot 2 = 52 + 43 + 2 \cdot 322, \quad 3 \cdot 22 = 52 + 322,$$

and these two equations imply the before-mentioned syzygy, $32 \cdot 2 - 3 \cdot 22 = 0$, in virtue of which the two reductions become equivalent; or say there remains a single equation serving for the reduction: the most simple form is $52 = 3 \cdot 22 - 322$.

Weight 8: there is only the seminvariant 53, and the product $33 \cdot 2$: this gives the reduction.

Weight 9: seminvariants 54, 522: products $32^2 \cdot 2$, $32 \cdot 22$, $3 \cdot 2^3$; there is between the first and last of these a syzygy; and this being satisfied, there remain two equations for the reductions.

Weight 10: seminvariants 55, 532: products $332 \cdot 2$, $33 \cdot 22$: these give the reductions.

Weight 11: seminvariants 533, 542, 5222: products $333 \cdot 2$; $32^3 \cdot 2$, $322 \cdot 2^2$, $32 \cdot 2^3$, $3 \cdot 2^4$.

34. Observe that the seminvariants, and in like manner the products, form two classes, according as the symbols contain three odd numbers or a single odd number: these correspond separately to each other, for the development of any product will contain only seminvariants having each of them as many odd numbers as there are odd numbers in the product. Hence for the weight 11 just referred to, $333 \cdot 2$ serves for the reduction of 533; $32^3 \cdot 2$, $322 \cdot 2^2$, $32 \cdot 2^3$, $3 \cdot 2^4$ are connected the first and fourth of them by a syzygy, and the second and third by a syzygy; and there remain two equations serving for the reduction of the two seminvariants 542, 5222.

It is easy to show in this manner that there is no quintic perpetuant for any weight under 15; and that there is a single quintic perpetuant for the weight 15.

35. Generally the syzygies exist only for an odd weight $w = 2\beta + 3$ between the products $32^{\beta-1} \cdot 2$, $32^{\beta-2} \cdot 2^2, \dots, 32 \cdot 2^{\beta-1}$, $3 \cdot 2^\beta$; viz. there is a syzygy between the first and last terms; a syzygy between the second and last but one terms; and so on.

The existence of these syzygies at once appears from the principle of decapitation: decapitating $32^{\beta-1} \cdot 2 - 3 \cdot 2^\beta$, we have $-2^{\beta-1} \cdot 2^{\beta-1}$, which is identically $= 0$, hence the function contains no term of the degree 5, that is, $32^{\beta-1} \cdot 2 - 3 \cdot 2^\beta = 0$; and similarly for the other pairs of terms.

There are β terms: hence in the case β even, we have $\frac{1}{2}\beta$ pairs of terms and therefore $\frac{1}{2}\beta$ syzygies: in the case β odd, there is a middle term, not connected by a syzygy with any other term, and the number of syzygies is thus $= \frac{1}{2}(\beta - 1)$: writing for β its value $\frac{1}{2}(w - 3)$, the number is $= \frac{1}{4}(w - 3)$ and $\frac{1}{4}(w - 5)$ in the two cases respectively: and it thus appears that the G.F. is $= \frac{x^7}{2 \cdot 4}$ as already mentioned.

36. In the case w an odd number, we have seminvariants, and in like manner products, containing respectively one odd number, three odd numbers, five odd numbers, and so on: thus $w = 15$, we have

<i>Seminvariants.</i>	<i>Products.</i>
555	$3332^2 \cdot 2$ 3 equations,
5532	$3332 \cdot 2^2$
5433	$333 \cdot 2^3$
5332 ²	
5442	$32^5 \cdot 2$ 6, = 3 equations;
542 ³	$32^4 \cdot 2^2$
52 ⁵	$32^3 \cdot 2^3$
	$32^2 \cdot 2^4$
	$32 \cdot 2^5$
	$3 \cdot 2^6$